nevertheless be spontaneously broken if none of these configurations are minima of the potential. In contrast, any supersymmetric field configuration gives the potential a value of zero, which is necessarily lower than the value of the potential for any non-supersymmetric configuration, so the existence of *any* supersymmetric field configuration insures that supersymmetry is unbroken. As we will see in Section 27.6, this conclusion goes beyond the tree approximation used in this section; it is unaffected by corrections of any finite order in perturbation theory.

It may appear that Eqs. (27.4.10) and (27.4.11) impose too many conditions on the scalar fields to expect a solution, without some fine-tuning of the superpotential. However, for a gauge group of dimensionality $D$, the superpotential $f(\phi)$ is subject to the $D$ constraints

$$\sum_m \frac{\partial f(\phi)}{\partial \phi_m}\Big(t_A\phi\Big)_m = 0\,, \tag{27.4.12}$$

for all $A$ and all $\phi$. Hence if $\phi$ has $N$ independent components, then the number of *independent* conditions (27.4.10) is $N-D$, while the number of conditions (27.4.11) is $D$, so there are just $N$ conditions altogether. With the number of conditions equal to the number of free variables, it is likely to find solutions for generic superpotentials. In fact, it is more usual to find solutions than not. For instance, for chiral scalar superfields in a non-trivial representation of a semi-simple gauge group we have $\xi_A = 0$, while $f(\phi)$ can have no terms linear in the $\phi_n$, so both Eqs. (27.4.10) and (27.4.11) are satisfied for $\phi_{n0} = 0$. There may be other solutions of Eqs. (27.4.10) and (27.4.11) which break the gauge symmetries, but in such a theory supersymmetry cannot be broken, at least not in the tree approximation, and, as we shall see in Section 27.6, not in any order of perturbation theory.

More generally, it is easy to see that, even if the gauge group has $U(1)$ factors and even if the superpotential involves gauge-invariant superfields, if there exists a set of scalar field values $\phi_{n0}$ that satisfy Eq. (27.4.10), then there is another that satisfies both Eq. (27.4.10) *and* Eq. (27.4.11), provided only that the Fayet–Iliopoulos constants $\xi_A$ all vanish. To show this, we note that since the superpotential $f(\phi)$ does not involve $\phi^*$, it is invariant not only under ordinary gauge transformations $\phi \to \exp(i\sum_A \Lambda_A t_A)\phi$ with $\Lambda_A$ arbitrary real numbers, but also under transformations with $\Lambda_A$ arbitrary complex numbers. Under all these transformations the $\mathscr{F}$-terms in Eq. (27.4.10) transform linearly, so if $\phi_0$ satisfies Eq. (27.4.10), then so does $\phi^\Lambda \equiv \exp(i\sum_A \Lambda_A t_A)\phi_0$. On the other hand, the scalar product $[\phi^\dagger\phi]$ is not invariant under transformations with $\Lambda_A$ complex, but $[\phi^{\Lambda\dagger}\phi^\Lambda]$ does remain real and positive for complex $\Lambda_A$, so it is bounded below, and therefore has a minimum. For $\xi_A = 0$, the condition that $[\phi^{\Lambda\dagger}\phi^\Lambda]$ be